SOFTWARE REQUIREMENTS SPECIFICATION

Domain: Digital Governance and E-Services

Problem Statement: Unified Citizen Service Delivery Platform (UCSDP)

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1. Preface

Expected Readership

This Software Requirements Specification (SRS) document has been prepared for everyone involved in bringing the Unified Citizen Service Delivery Platform (UCSDP) to life—from faculty reviewers and project stakeholders to system analysts, designers, developers, and quality assurance teams working within the digital governance and e-services space. We've written this document in straightforward, accessible language to ensure that readers from all backgrounds, whether deeply technical or not, can grasp the system's requirements and vision.

Summary

Since this is our first version, there aren't any changes from previous iterations to report. Moving forward, we anticipate that future versions will reflect updates driven by stakeholder input, shifts in government policies, emerging technologies, or natural evolution in what the system needs to accomplish.

2. Introduction

2.1. Need for the System

Right now, government services are scattered across numerous departments and platforms, which creates real headaches for everyday citizens trying to get things done. People face long wait times, lack of clarity about where their applications stand, the frustration of submitting the same documents multiple times, and the inconvenience of having to physically show up at government offices just to complete simple tasks.

The Unified Citizen Service Delivery Platform (UCSDP) exists to solve these problems. We're building a single digital hub where citizens can access a wide range of government services in a way that's convenient, transparent, and well-organized—all from the comfort of their homes or local service centers.

2.2. System Overview and Functions

The UCSDP serves as a centralized digital platform designed to bring essential government services online. The system accommodates various user types and delivers services including:

- Applying for and renewing official certificates (such as income, caste, and residential certificates) entirely online
- Managing welfare programs, subsidies, and scholarships with built-in eligibility verification
- Handling permissions and authorizations for public events and vendor activities
- Updating and correcting official records
- Submitting and resolving grievances or complaints
- Tracking application progress and receiving timely notifications.

The platform ensures smooth interactions between citizens and government personnel through well-defined workflows and role-specific access controls.

2.3. Interaction with Other Systems

The UCSDP won't work in isolation—it will coordinate with various supporting systems to deliver services effectively. These supporting systems include:

- Citizen record and identity databases for verification purposes
- Payment interfaces to handle fee collection (in a simulated environment for now) These connections will enable accurate identity verification, prompt communication, and consistent data handling across all services offered.

2.4. Business and Strategic Objectives

The UCSDP directly supports the digital governance goals of the organization commissioning it. Specifically, the system aims to:

- Make government services more accessible and efficient for everyone
- Cut down on manual paperwork and eliminate administrative bottlenecks
- Increase transparency and accountability in how services are delivered
- Foster a more citizen-focused and inclusive approach to governance

By meeting these objectives, the UCSDP plays a vital role in the broader digital transformation journey and helps improve how public services are managed over the long term.

3. Glossary

Here are the key terms used throughout this document:

Term	Definition
Unified Citizen Service Delivery Platform (UCSDP)	A web-based system that brings together multiple government services on a single online platform, making them easily accessible to citizens.
Digital Governance	The practice of using digital technologies by government organizations to deliver services and engage with citizens more effectively.

E-Services	Government services provided electronically through the internet, eliminating the need for in-person visits to physical offices.
Citizen	An individual who uses the system to submit applications for government services or to report issues and complaints.
Department Officer (First-Level Officer)	A government official tasked with reviewing and processing citizen applications as the first point of official evaluation.
Department Supervisor (Second-Level Officer)	A senior official who oversees service processing, manages escalated situations, and has the authority to override decisions when necessary.
Grievance	A formal complaint or concern raised by a citizen regarding service delays, application rejections, payment issues, or overall service quality.
Grievance Officer	An official specifically designated to receive, investigate, and work toward resolving grievances submitted by citizens.
Authentication	The process of confirming a user's identity before granting them access to the system.
SLA (Service Level Agreement)	A specified time frame within which a particular service or grievance must be processed and completed.

4. User Requirements Definition

This section outlines the services and capabilities that the system will provide to each type of user.

4.1 User Services (Functional Requirements)

4.1.1 Citizen

- The system will enable citizens to submit applications for various government services and certificates.
- The system will allow citizens to upload and manage the documents they need to support their applications.
- The system will let citizens monitor the status of their applications in real-time.
- The system will provide citizens with a way to raise grievances or complaints related to any service they've received.

- The system will enable citizens to request corrections to their official records.

4.1.2 Department Officer (First-Level Officer)

- The system will allow officers to carefully review and verify applications that have been submitted.
- The system will give officers the ability to approve or reject applications based on their assessment.
- The system will enable officers to request additional information or clarifications from citizens when needed.

4.1.3 Department Supervisor (Second-Level Officer)

- The system will allow supervisors to keep track of pending and overdue applications across their department.
- The system will enable supervisors to step in and manage escalated cases, including the ability to override decisions when circumstances warrant it.

4.1.4 Grievance Officer

- The system will allow grievance officers to receive and begin processing citizen complaints.
- The system will enable grievance officers to resolve complaints directly or escalate them to higher authorities when necessary.

4.2. Non-Functional User Requirements

- The system will be intuitive and user-friendly, requiring minimal training for people to get started.
- The system will respond to user actions promptly, keeping wait times within reasonable limits.
- The system will maintain secure handling and storage of all user data.
- The system will remain accessible to users around the clock, with exceptions only for scheduled maintenance periods.

4.3. Product and Process Standards

- The system will adhere to recognized best practices for web application development.
- The system will comply with fundamental digital governance guidelines established for public services.
- User data will be managed in accordance with prevailing data privacy and security standards.

5. System Requirements Specification

5.1 Functional Requirements

- **5.1.1** The system will provide a secure registration and login process for all users.
- **5.1.2** The system will enable citizens to submit service applications.
- **5.1.3** The system will support uploading, verifying, and securely storing documents.
- **5.1.4** The system will give department officers the ability to approve, reject, or ask for additional information regarding applications.
- **5.1.5** The system will facilitate the complete grievance lifecycle, including registration, processing, escalation, and closure.
- **5.1.6** The system will track the status of applications and grievances according to pre-defined workflows.
- **5.1.7** The system will allow supervisors to oversee and intervene in escalated cases as needed.

5.2 Non-Functional Requirements

- **5.2.1 Performance:** The system will deliver responses to user actions within timeframes that users find acceptable.
- **5.2.2 Security:** The system will guarantee secure access controls and robust protection of user data.
- **5.2.3 Availability:** The system will be accessible to users at all times, with brief interruptions only during scheduled maintenance windows.
- **5.2.4 Reliability:** The system will manage failures gracefully and maintain data consistency even under adverse conditions.
- **5.2.5 Usability:** The system will feature an intuitive design that requires minimal training for effective use.

5.3 External Interfaces

- **5.3.1 User Interface:** The system will offer a web-based interface that works seamlessly with standard web browsers.

- **5.3.2 Notification Interface:** The system will connect with email and SMS services to keep users informed about important updates.
- **5.3.3 Database Interface:** The system will interact with a centralized database to store and retrieve all relevant system data efficiently.

6. System Models

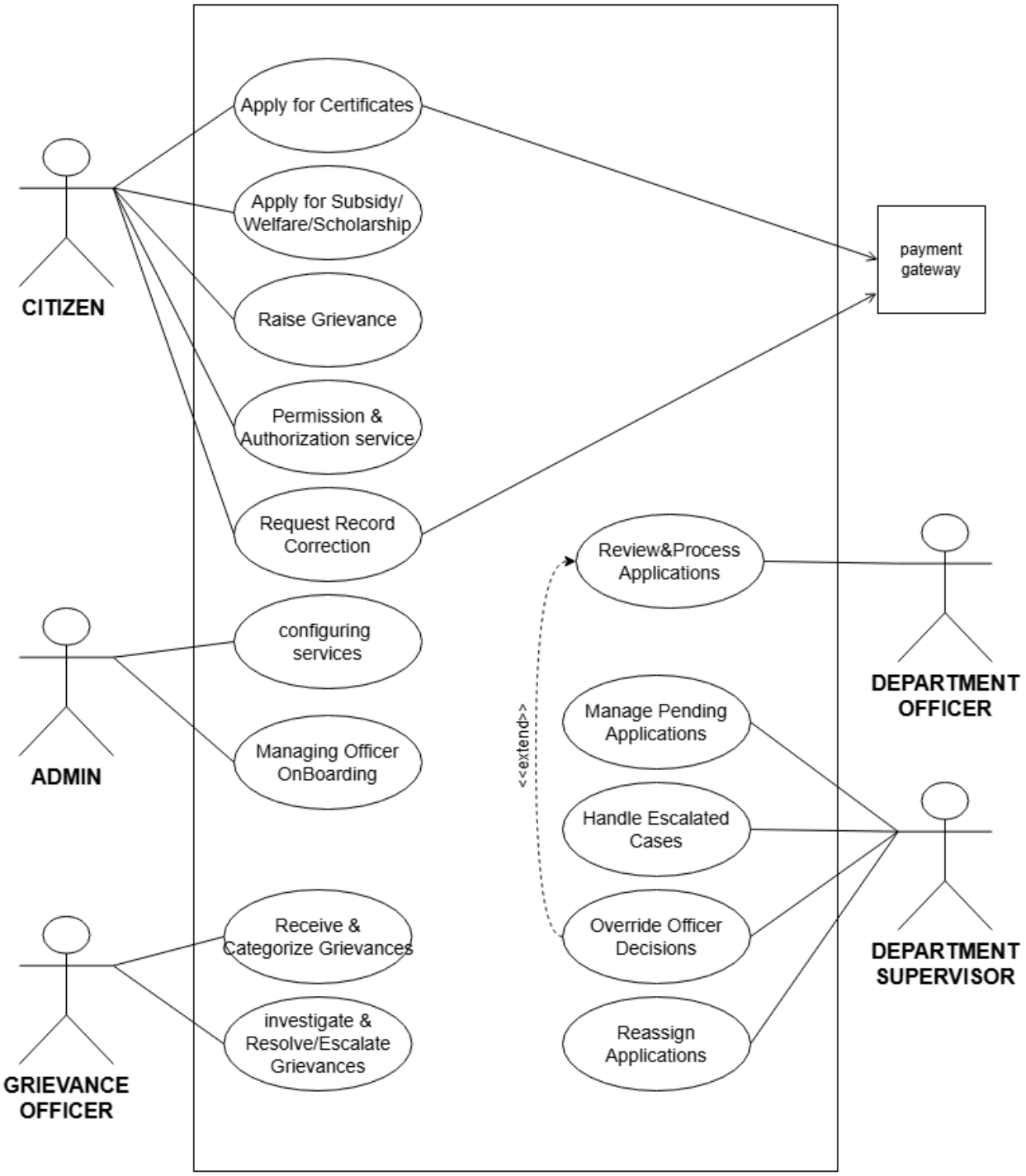
This section includes various diagrams and models that help visualize how the system functions:

- Use case diagrams
- Activity diagram
- Sequence diagrams

Use case diagram

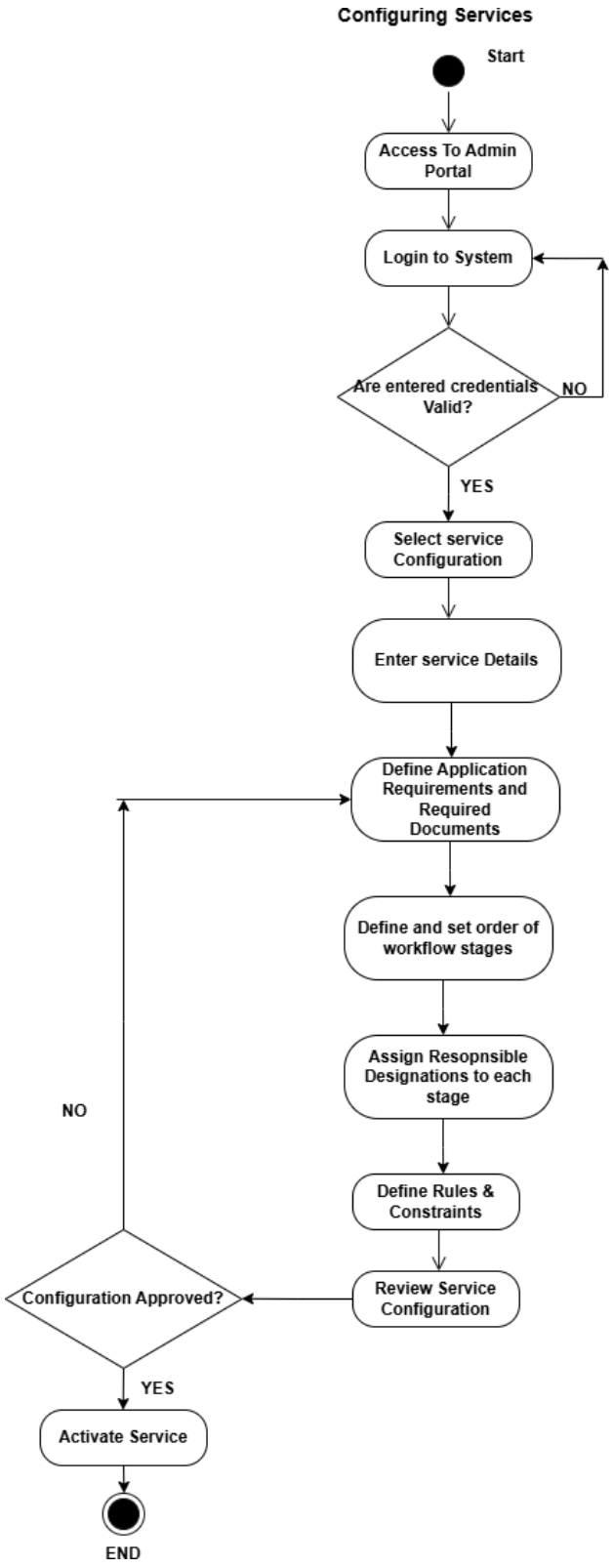
UNIFIED CITIZEN SERVICE DELIVERY PLATFORM

E-Governance System

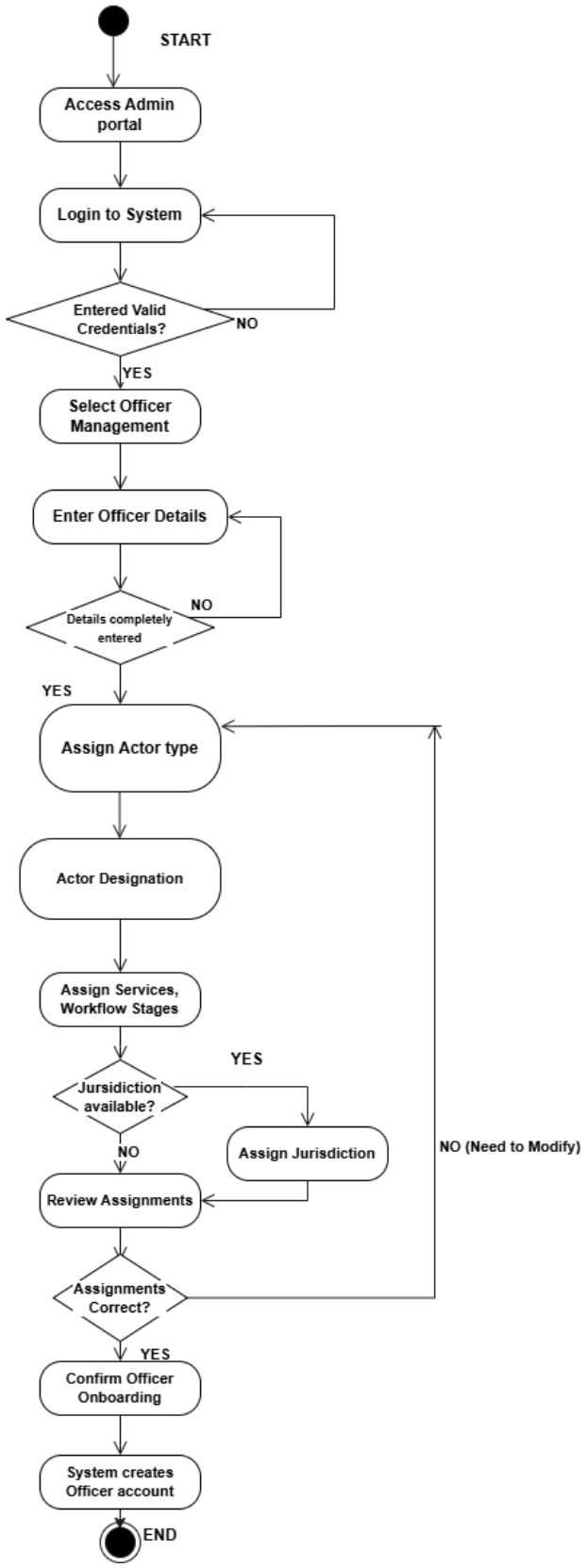


Activity diagrams

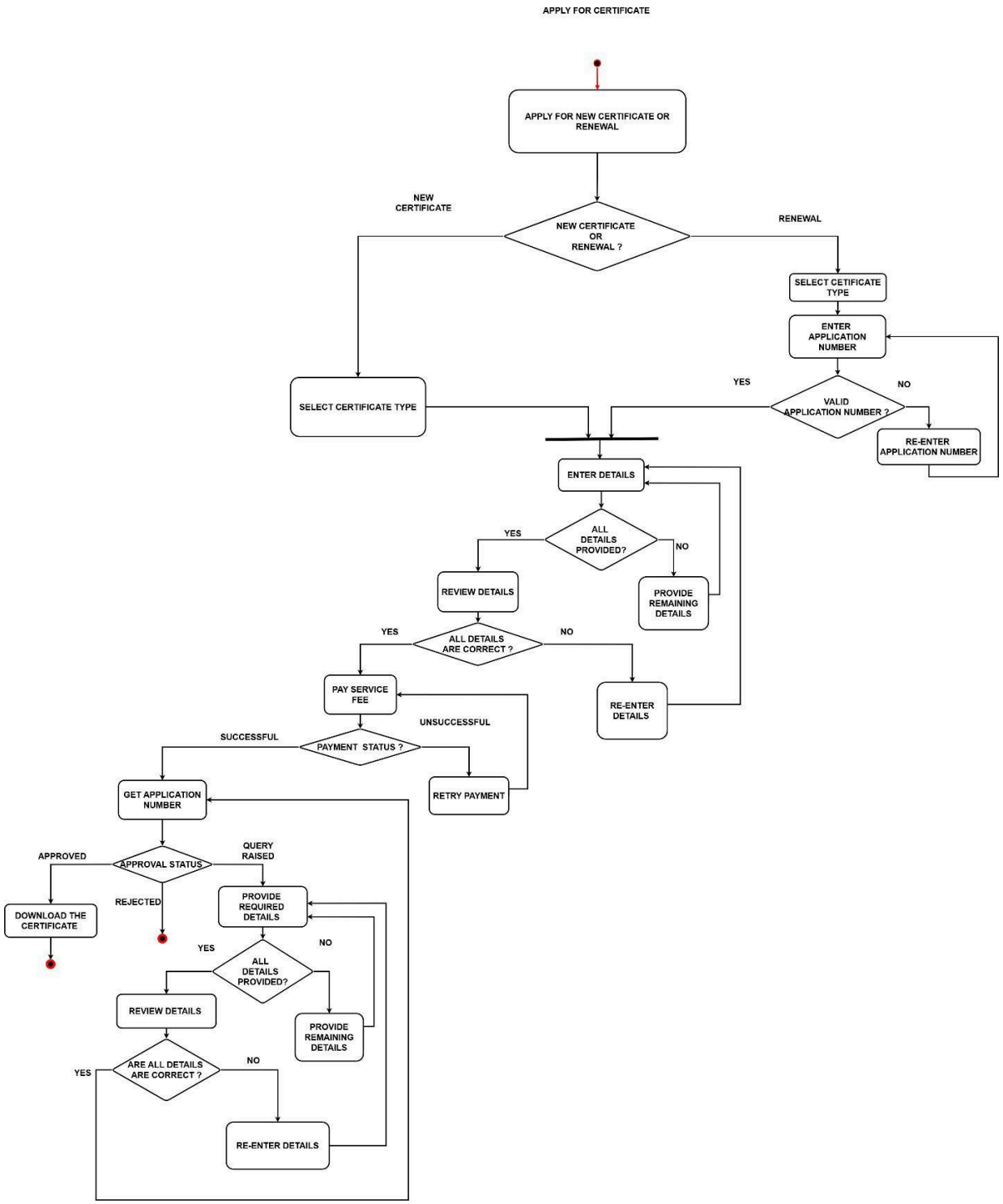
1.Admin


Configuring Services
Start
Access To Admin Portal
Login to System
Are entered credentials Valid?
NO
YES
Select service Configuration
Enter service Details
Define Application Requirements and Required Documents
Define and set order of workflow stages
Assign Resopnsible Designations to each stage
Define Rules & Constraints
Review Service Configuration
Configuration Approved?
NO
YES
Activate Service
END

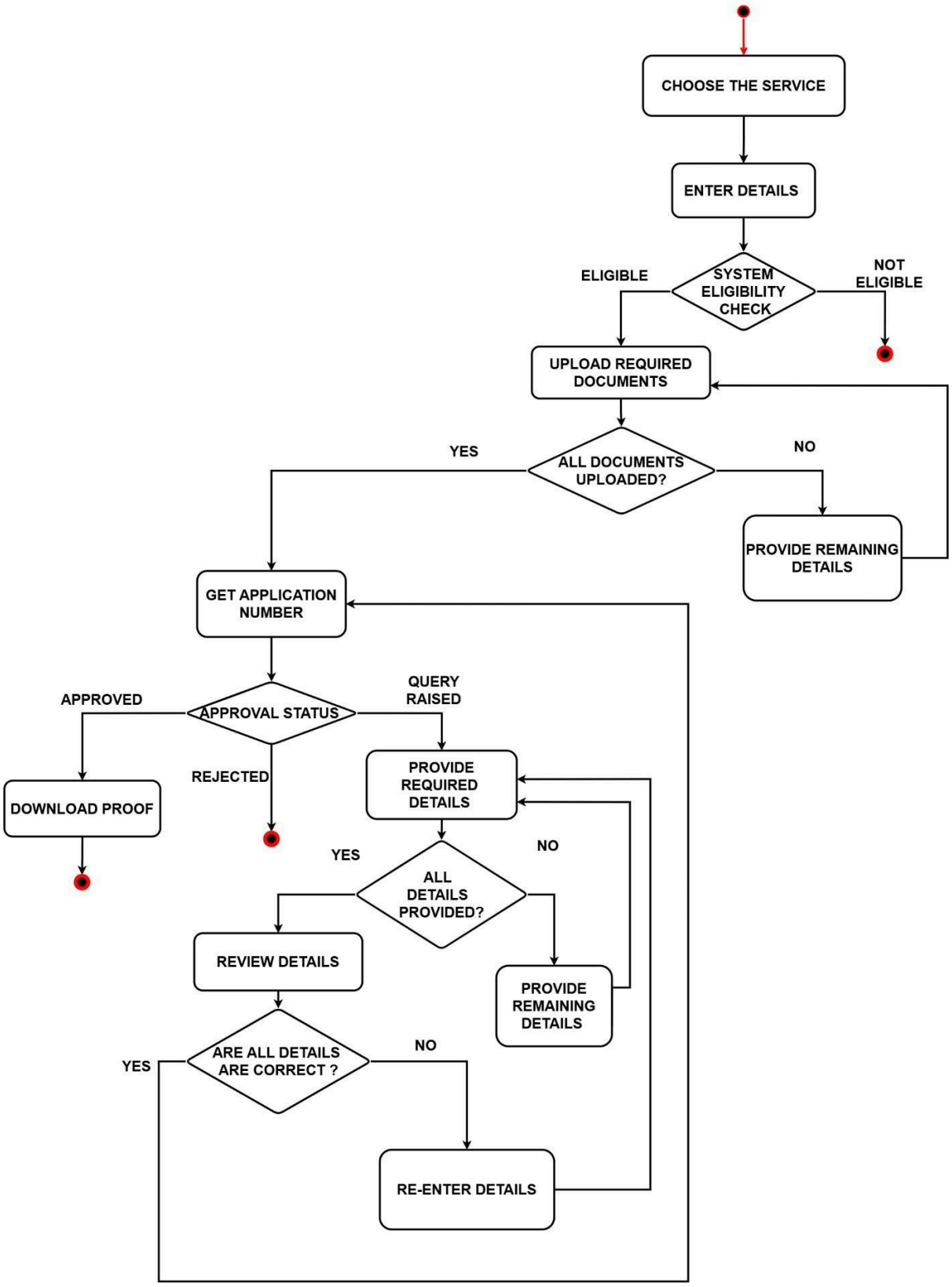
MANAGE OFFICER ONBOARDING



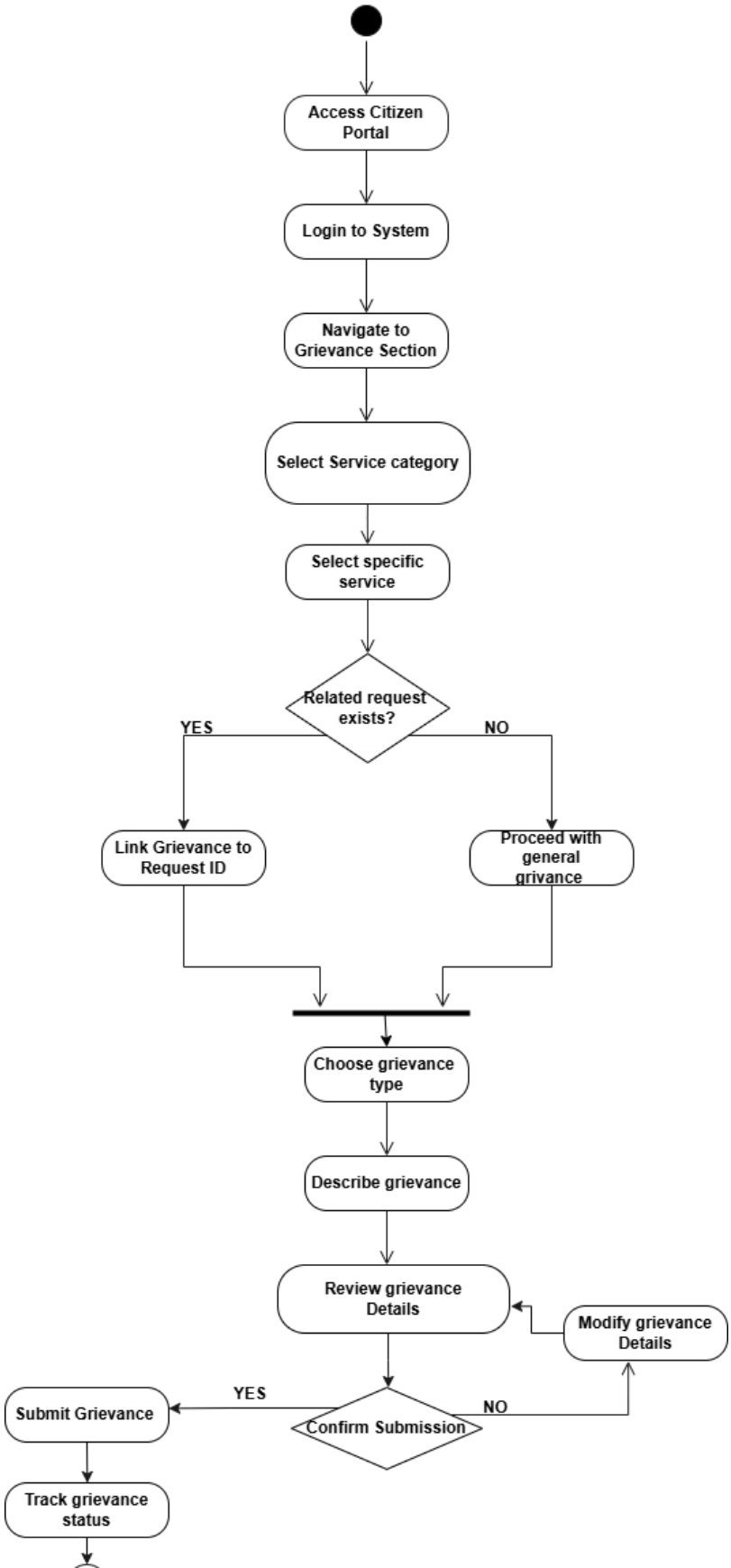
2.Citizen


APPLY FOR CERTIFICATE
APPLY FOR NEW CERTIFICATE OR RENEWAL
NEW CERTIFICATE
NEW CERTIFICATE OR RENEWAL ?
RENEWAL
SELECT CETIFICATE TYPE
ENTER APPLICATION NUMBER
YES
VALID APPLICATION NUMBER ?
NO
RE-ENTER APPLICATION NUMBER
SELECT CERTIFICATE TYPE
ENTER DETAILS
YES
ALL DETAILS PROVIDED?
NO
REVIEW DETAILS
PROVIDE REMAINING DETAILS
YES
ALL DETAILS ARE CORRECT ?
NO
PAY SERVICE FEE
RE-ENTER DETAILS
UNSUCCESSFUL
SUCCESSFUL
PAYMENT STATUS ?
RETRY PAYMENT
GET APPLICATION NUMBER
APPROVED
APPROVAL STATUS
QUERY RAISED
DOWNLOAD THE CERTIFICATE
REJECTED
PROVIDE REQUIRED DETAILS
YES
NO
ALL DETAILS PROVIDED?
REVIEW DETAILS
PROVIDE REMAINING DETAILS
YES
ARE ALL DETAILS ARE CORRECT ?
NO
RE-ENTER DETAILS

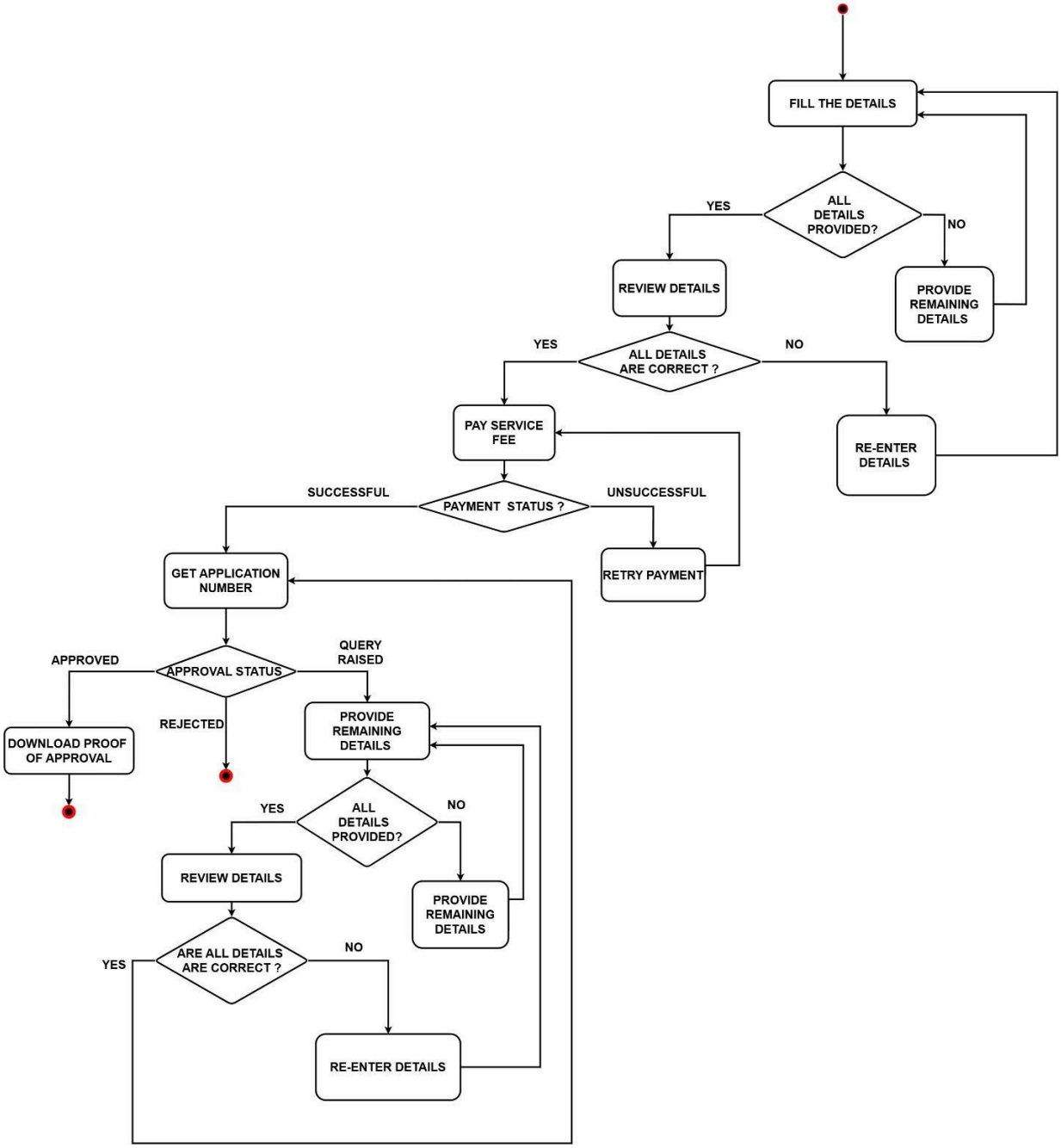
APPLY FOR SUBSIDY / WELFARE / SCHORLARSHIP



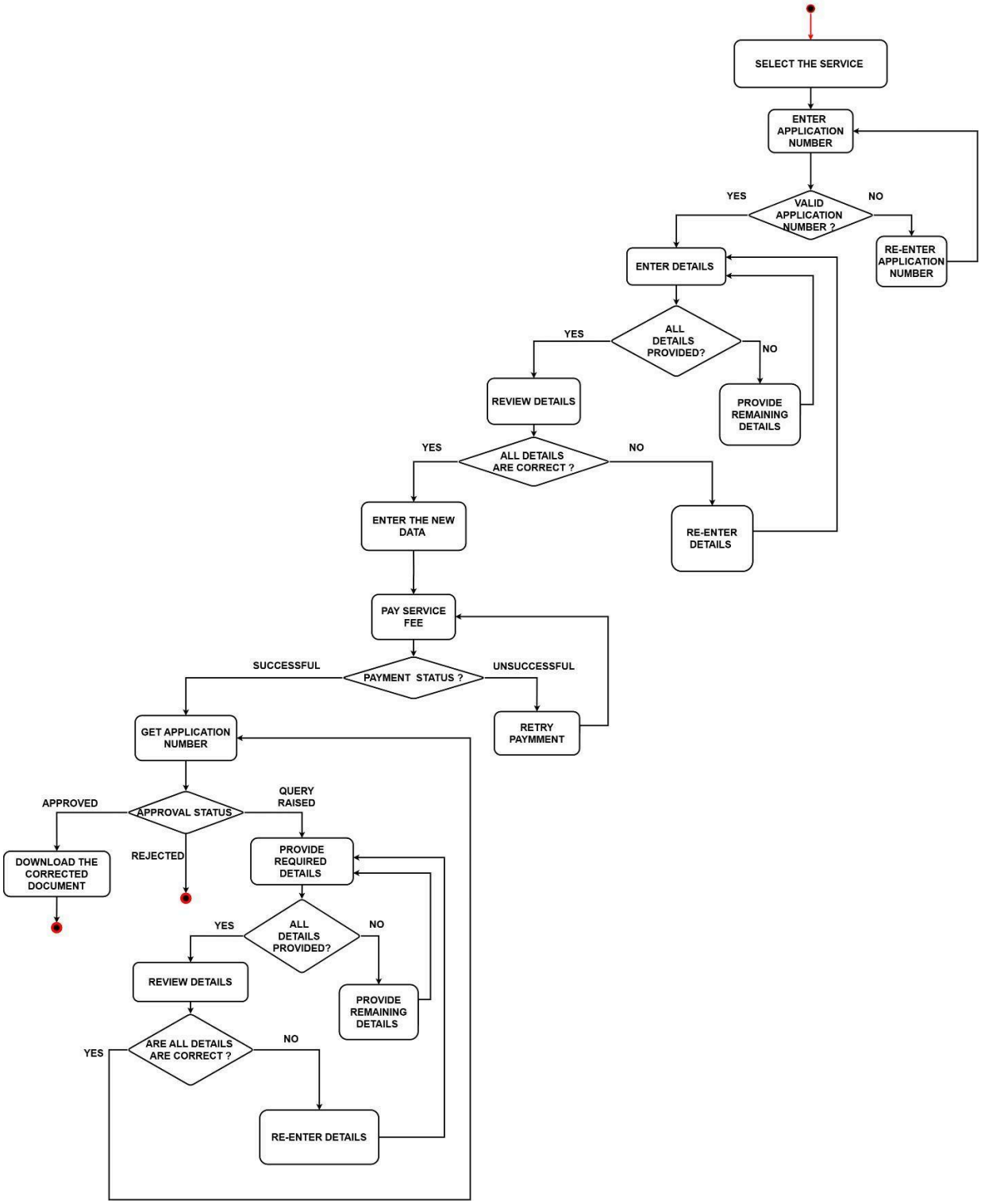
RAISE GRIEVANCE



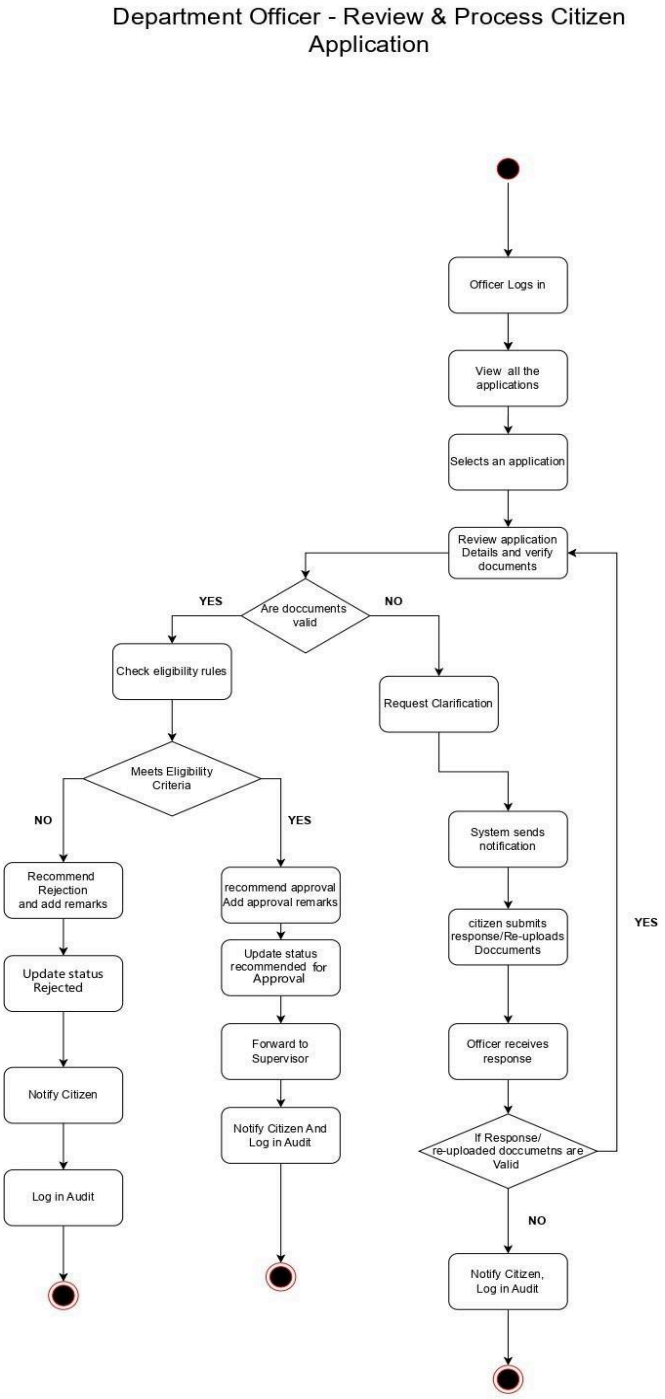
PERMISSION AND AUTHORIZATION SERVICE



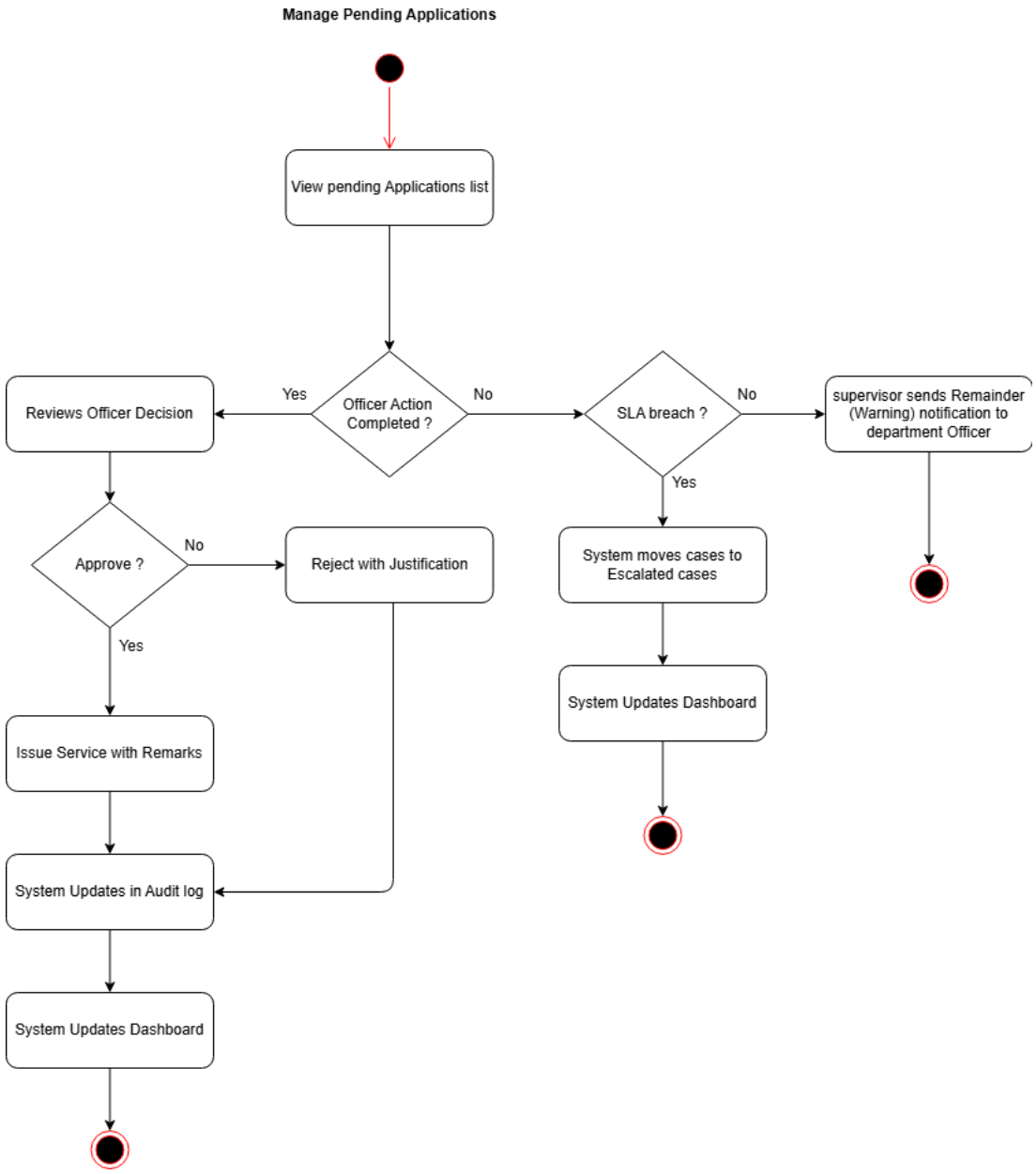
REQUEST RECORD CORRECTION

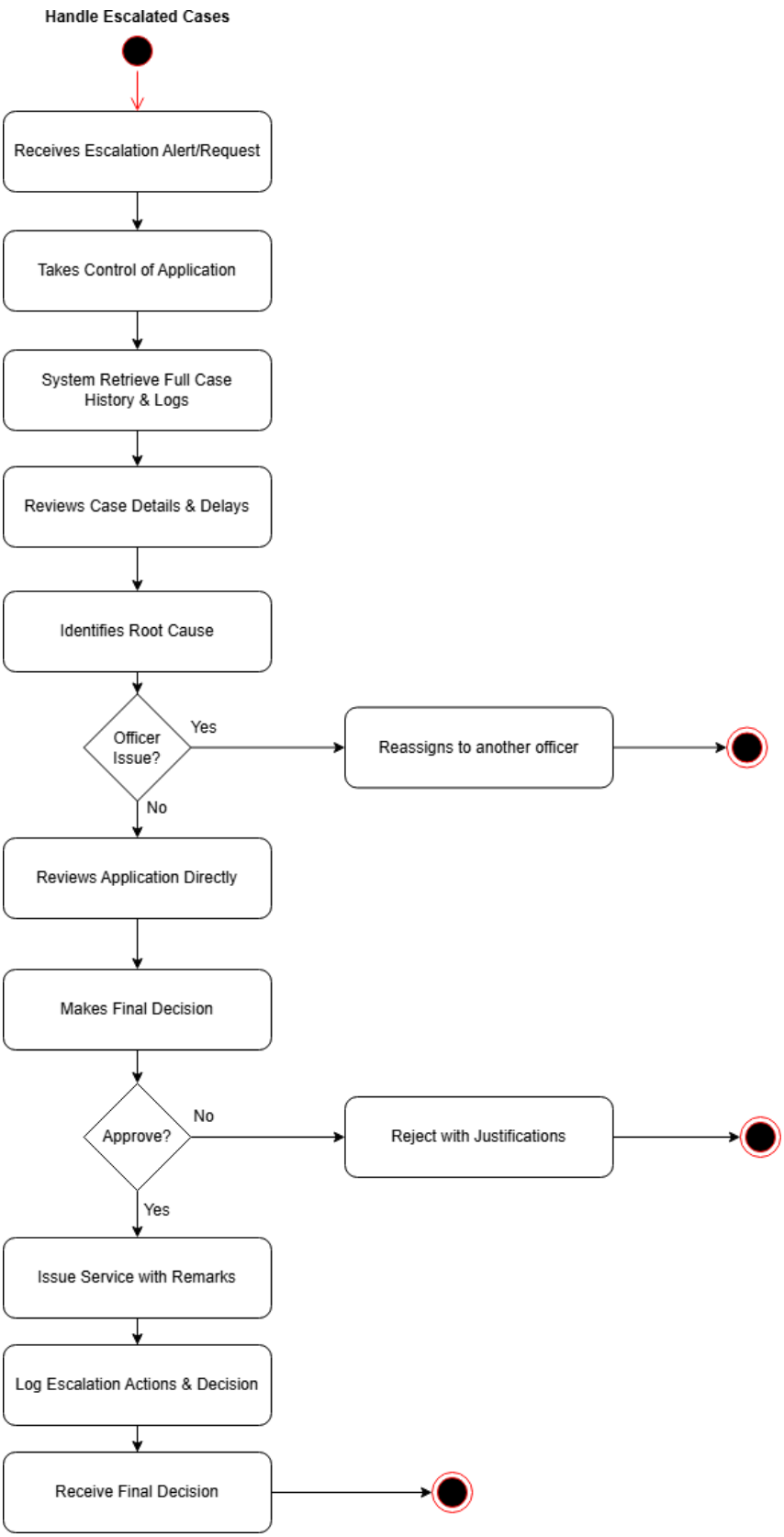


3. Department Officer

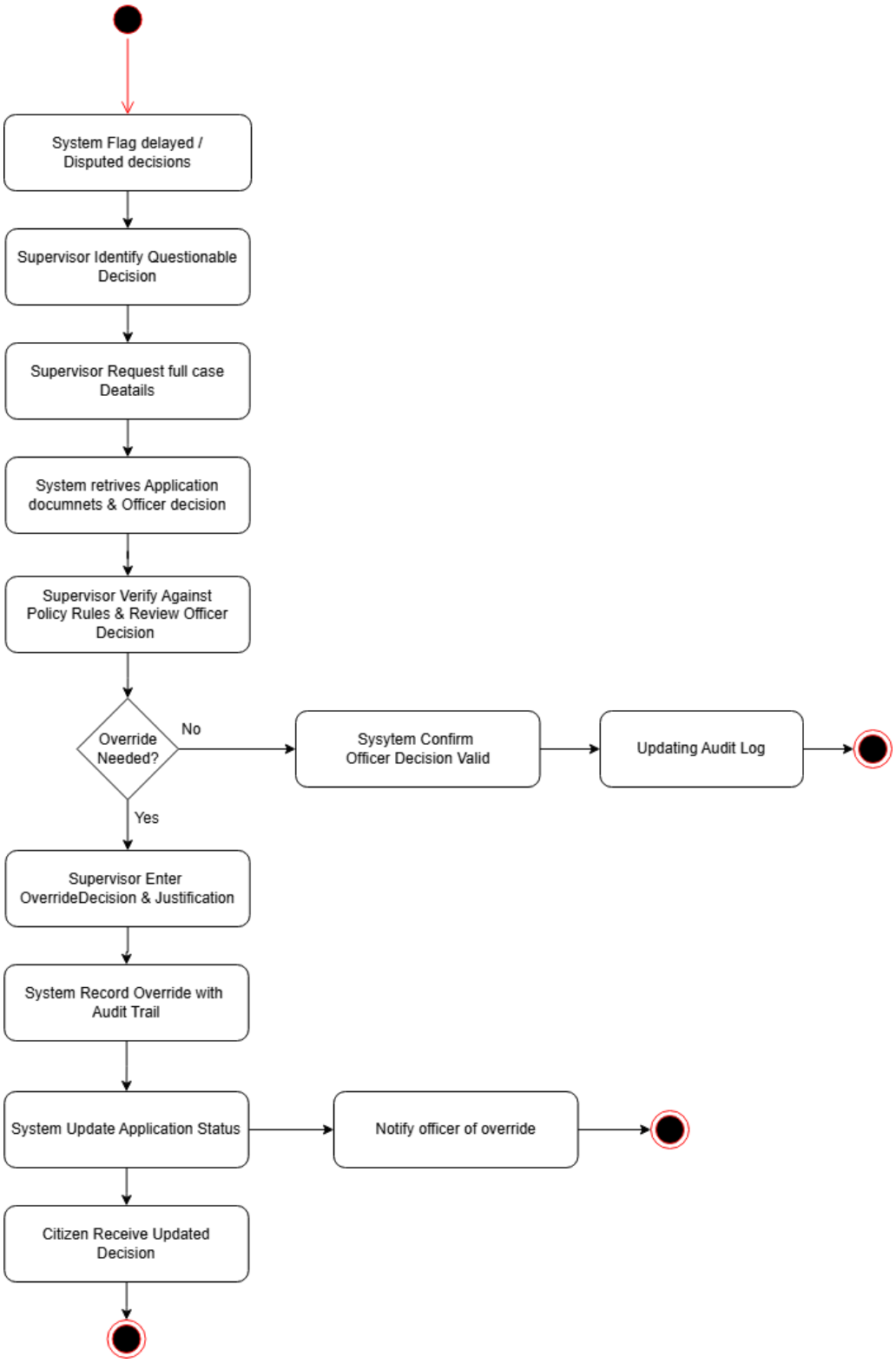


4. Department Supervisor

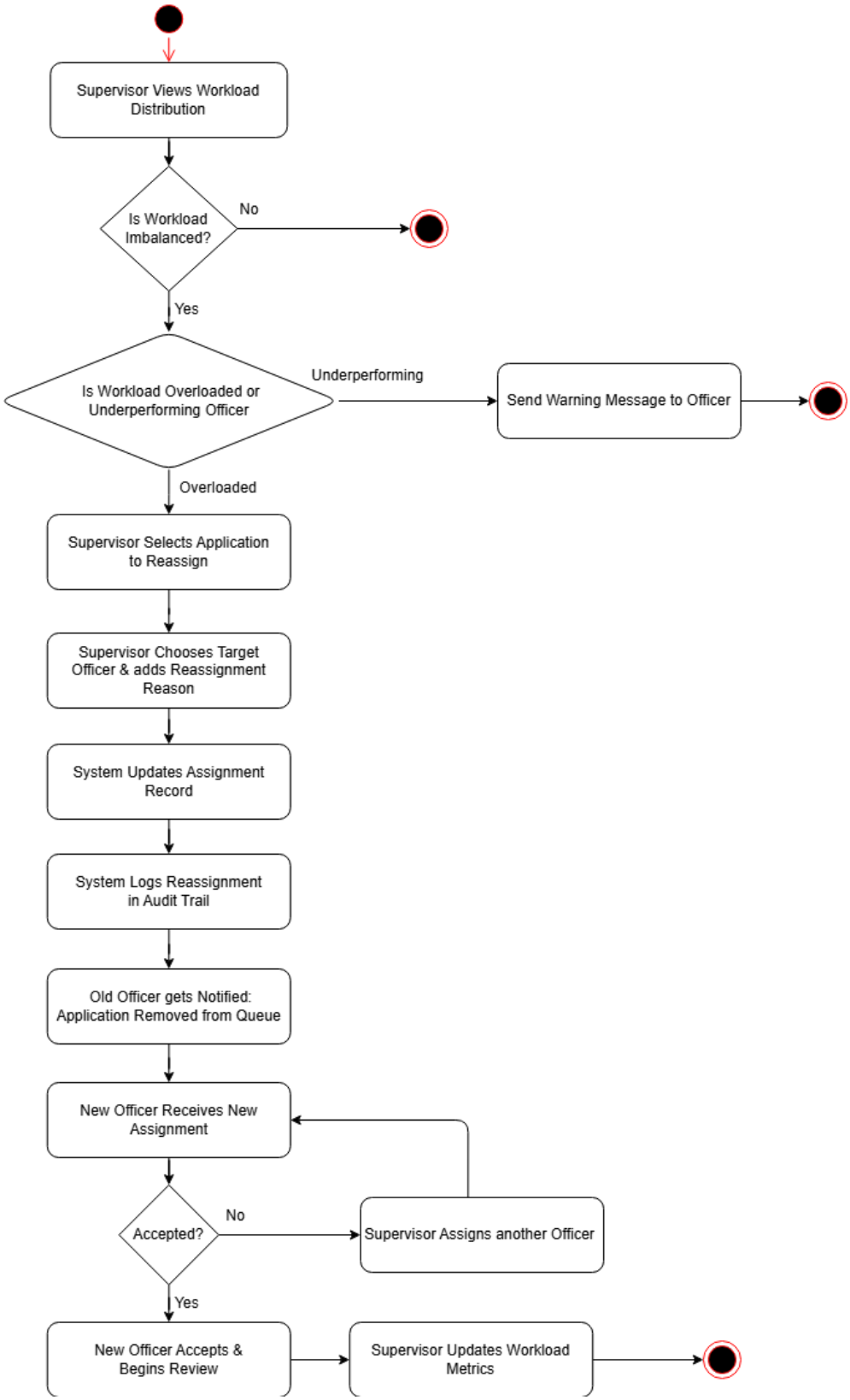



Handle Escalated Cases
Receives Escalation Alert/Request
Takes Control of Application
System Retrieve Full Case History & Logs
Reviews Case Details & Delays
Identifies Root Cause
Officer Issue?
Yes
Reassigns to another officer
No
Reviews Application Directly
Makes Final Decision
Approve?
No
Reject with Justifications
Yes
Issue Service with Remarks
Log Escalation Actions & Decision
Receive Final Decision

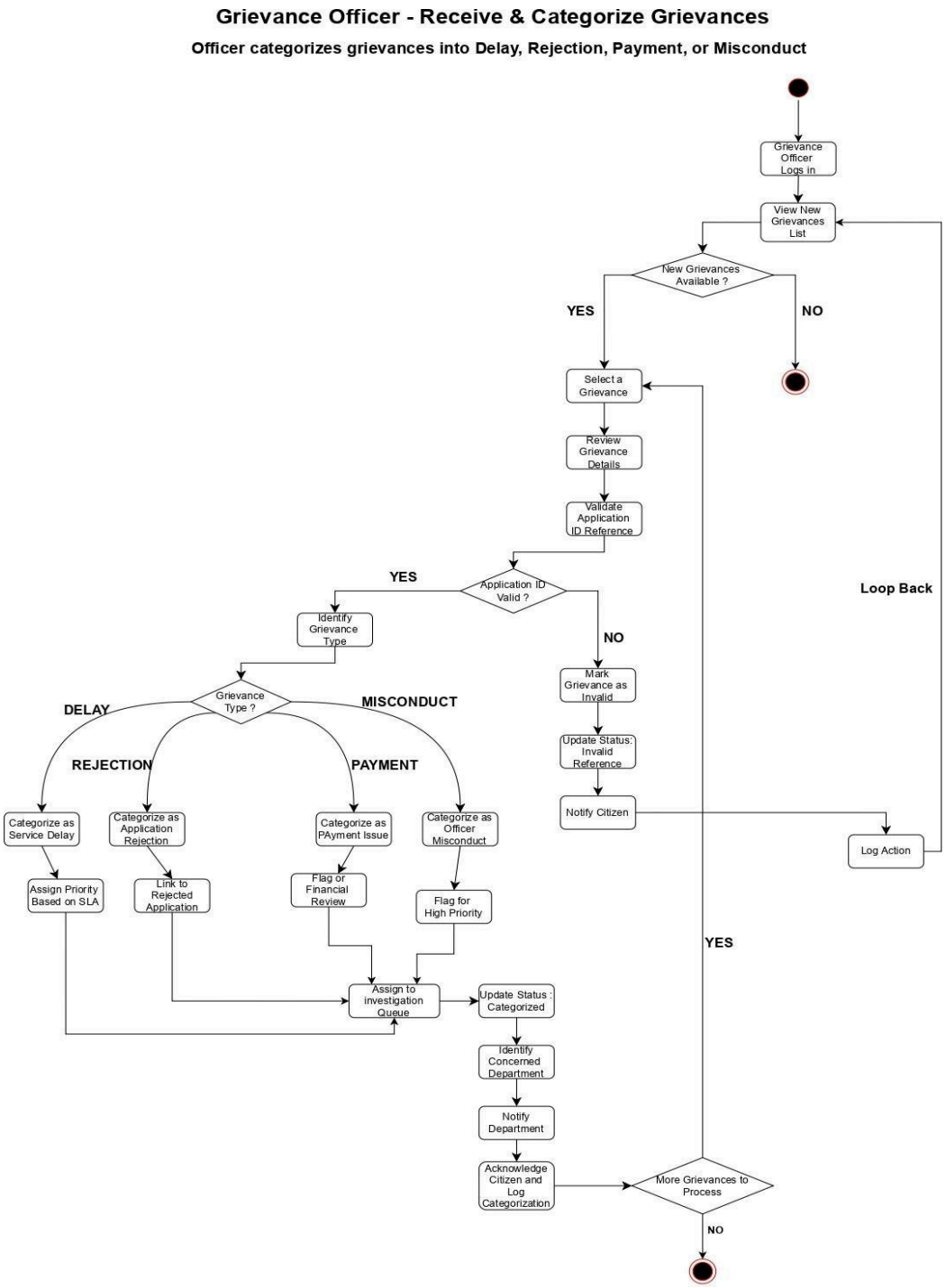
Override Officer Decisions



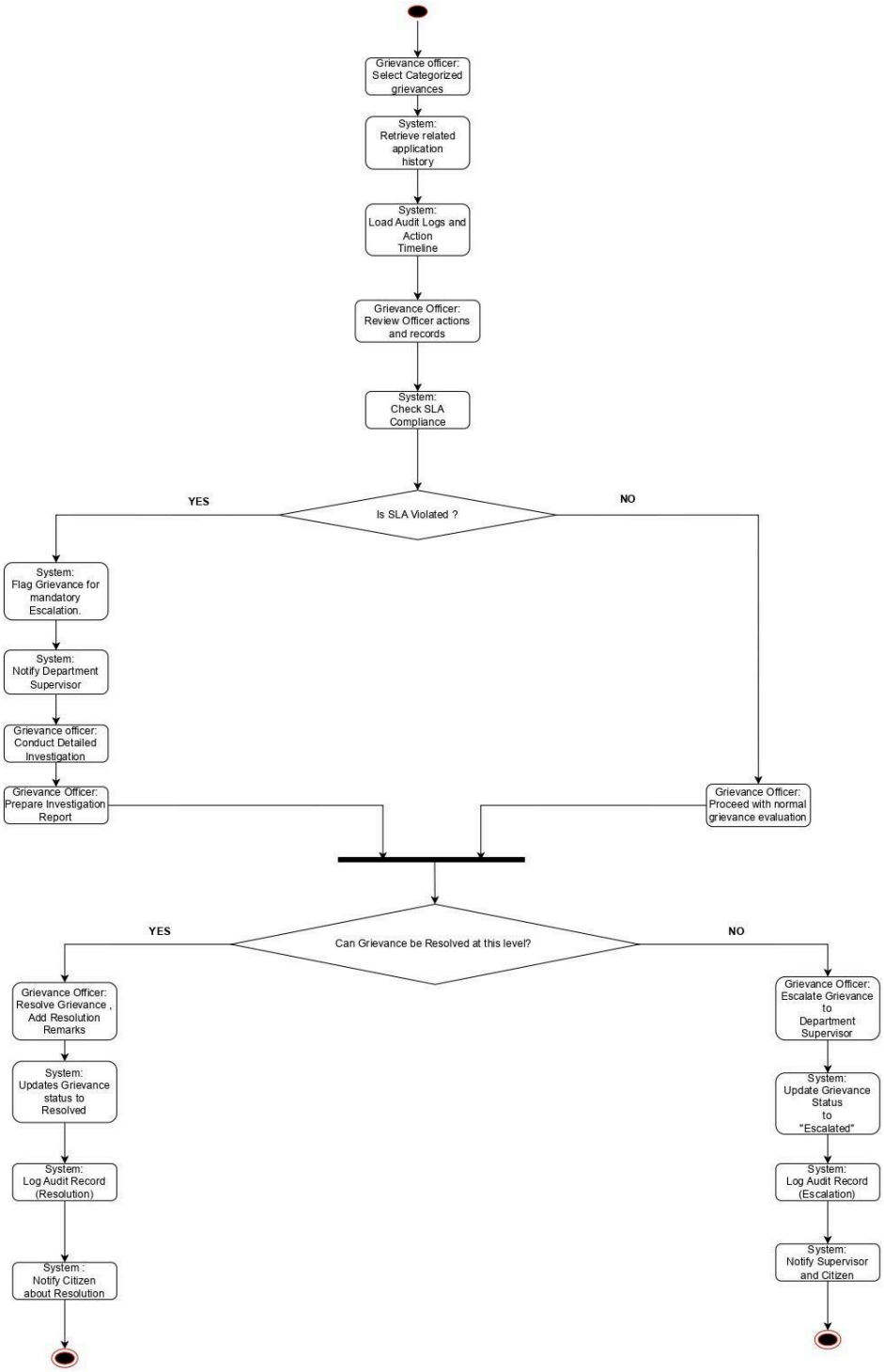
Reassign Applications



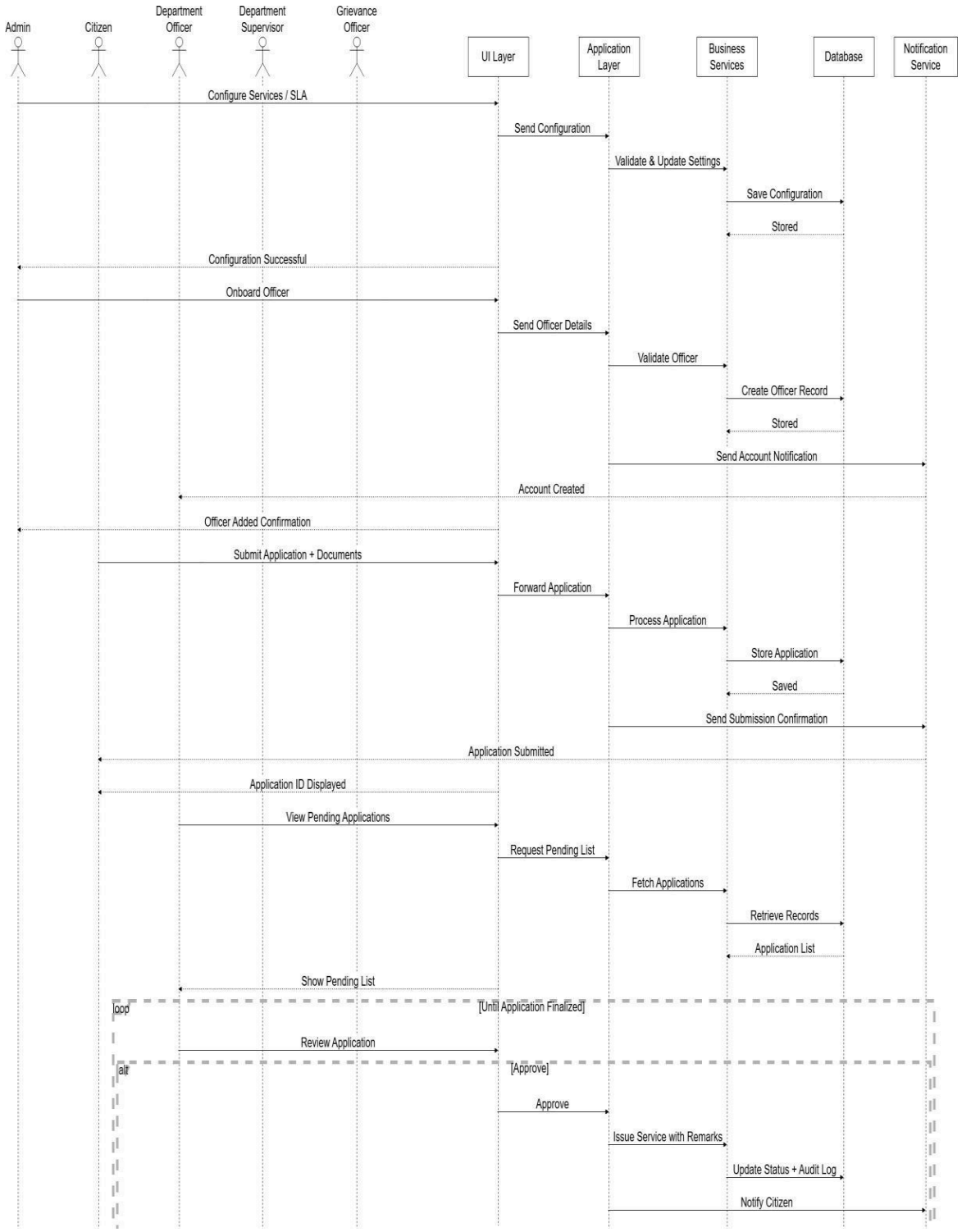
5. Grievance Officer

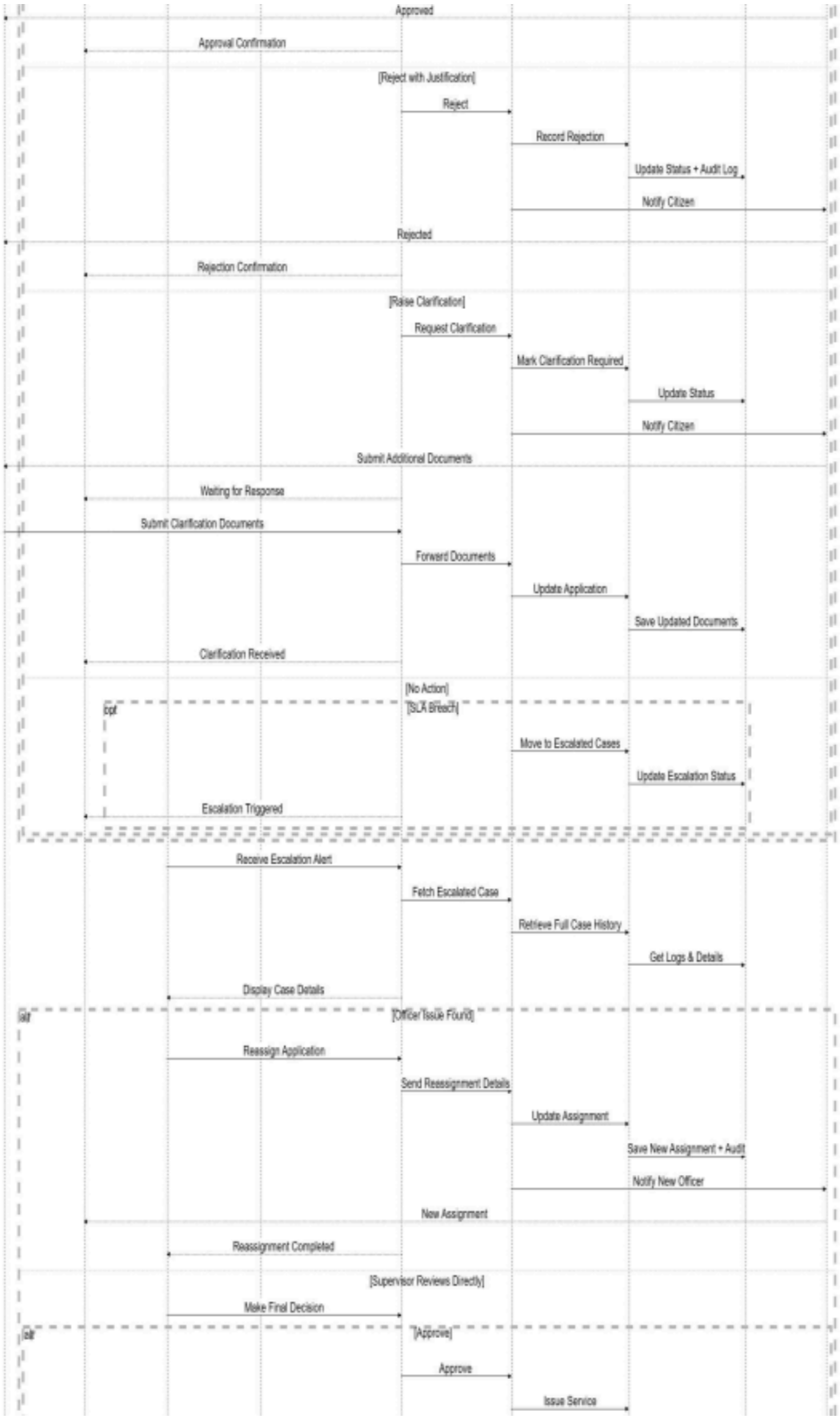


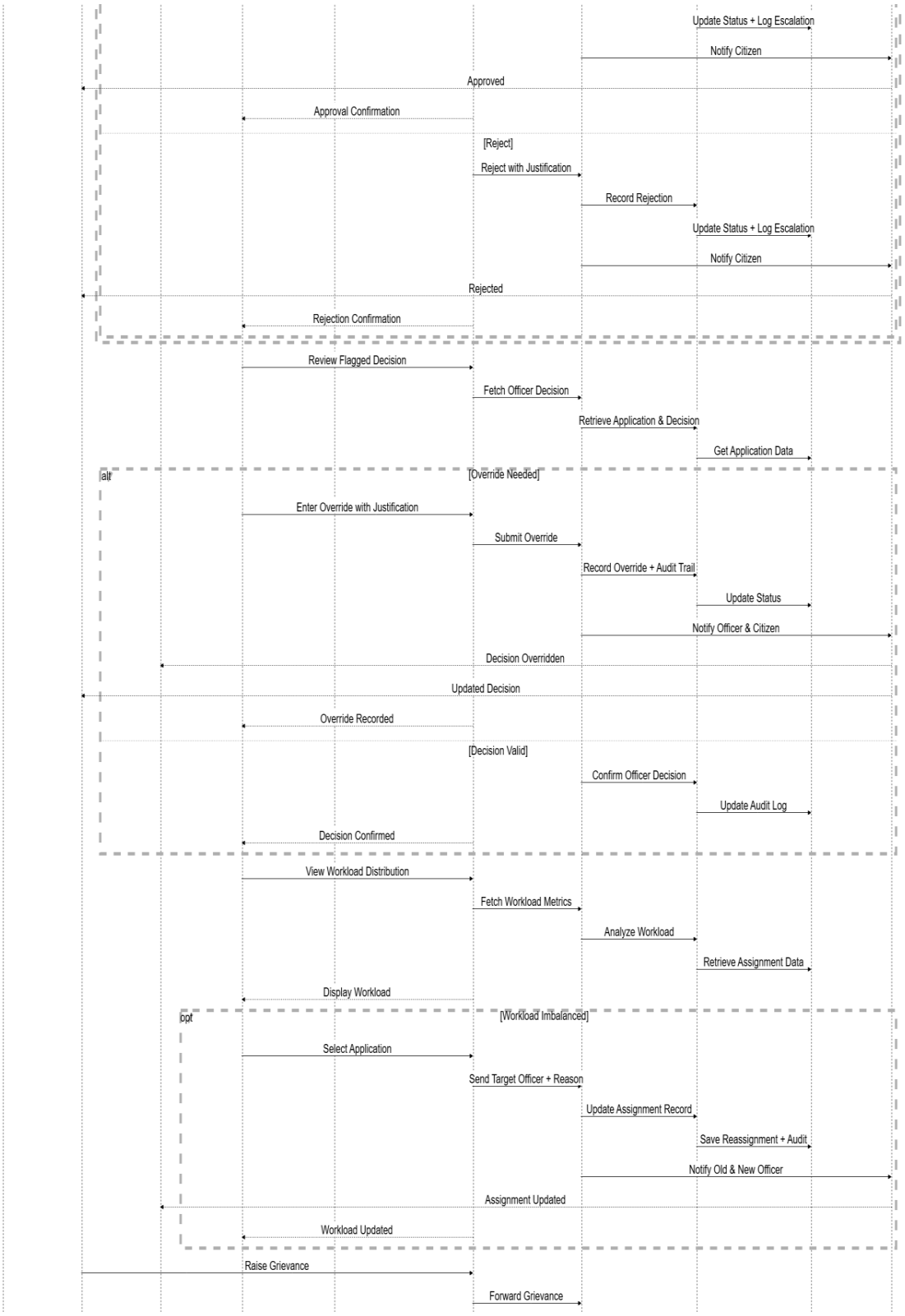
Grievance Officer – Investigate & Resolve / Escalate Grievance

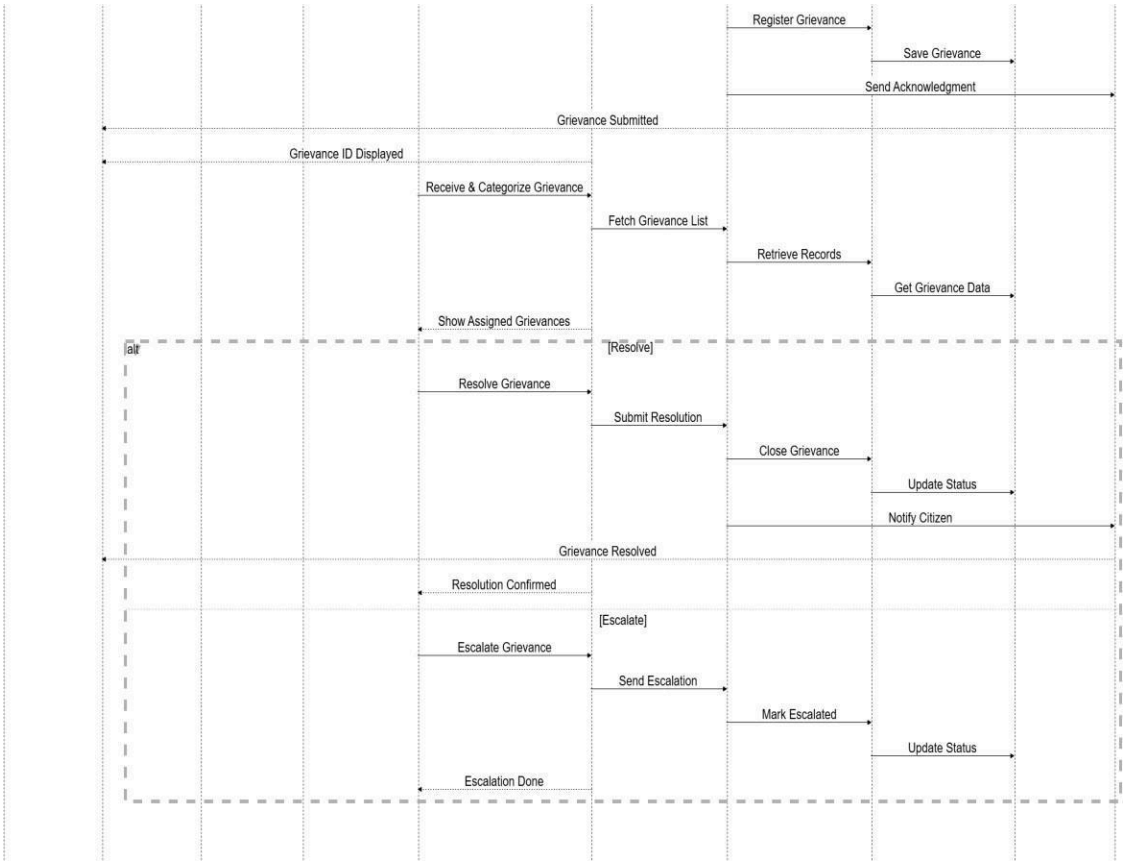


Sequence diagram


Admin
Citizen
Department Officer
Department Supervisor
Grievance Officer
UI Layer
Application Layer
Business Services
Database
Notification Service
Configure Services / SLA
Send Configuration
Validate & Update Settings
Save Configuration
Stored
Configuration Successful
Onboard Officer
Send Officer Details
Validate Officer
Create Officer Record
Stored
Send Account Notification
Account Created
Officer Added Confirmation
Submit Application + Documents
Forward Application
Process Application
Store Application
Saved
Send Submission Confirmation
Application Submitted
Application ID Displayed
View Pending Applications
Request Pending List
Fetch Applications
Retrieve Records
Application List
Show Pending List
loop
[Until Application Finalized]
Review Application
alt
[Approve]
Approve
Issue Service with Remarks
Update Status + Audit Log
Notify Citizen


Approved
Approval Confirmation
[Reject with Justification]
Reject
Record Rejection
Update Status + Audit Log
Notify Citizen
Rejected
Rejection Confirmation
[Raise Clarification]
Request Clarification
Mark Clarification Required
Update Status
Notify Citizen
Submit Additional Documents
Waiting for Response
Submit Clarification Documents
Forward Documents
Update Application
Save Updated Documents
Clarification Received
[No Action]
opt
[SLA Breach]
Move to Escalated Cases
Update Escalation Status
Escalation Triggered
Receive Escalation Alert
Fetch Escalated Case
Retrieve Full Case History
Get Logs & Details
Display Case Details
alt
[Officer Issue Found]
Reassign Application
Send Reassignment Details
Update Assignment
Save New Assignment + Audit
Notify New Officer
New Assignment
Reassignment Completed
[Supervisor Reviews Directly]
Make Final Decision
alt
[Approve]
Approve
Issue Service


Update Status + Log Escalation
Notify Citizen
Approved
Approval Confirmation
[Reject]
Reject with Justification
Record Rejection
Update Status + Log Escalation
Notify Citizen
Rejected
Rejection Confirmation
Review Flagged Decision
Fetch Officer Decision
Retrieve Application & Decision
Get Application Data
alt
[Override Needed]
Enter Override with Justification
Submit Override
Record Override + Audit Trail
Update Status
Notify Officer & Citizen
Decision Overridden
Updated Decision
Override Recorded
[Decision Valid]
Confirm Officer Decision
Update Audit Log
Decision Confirmed
View Workload Distribution
Fetch Workload Metrics
Analyze Workload
Retrieve Assignment Data
Display Workload
opt
[Workload Imbalanced]
Select Application
Send Target Officer + Reason
Update Assignment Record
Save Reassignment + Audit
Notify Old & New Officer
Assignment Updated
Workload Updated
Raise Grievance
Forward Grievance


Register Grievance
Save Grievance
Send Acknowledgment
Grievance Submitted
Grievance ID Displayed
Receive & Categorize Grievance
Fetch Grievance List
Retrieve Records
Get Grievance Data
Show Assigned Grievances
alt
[Resolve]
Resolve Grievance
Submit Resolution
Close Grievance
Update Status
Notify Citizen
Grievance Resolved
Resolution Confirmed
[Escalate]
Escalate Grievance
Send Escalation
Mark Escalated
Update Status
Escalation Done